

Data Reduction Strategy using Neural Adaptation Phenomenon in Internet of Things

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Abstract—Internet of Things (IoT) have got significant popularity among the researchers' community as they have been applied in numerous application domains. Most of the IoT applications are implemented with the help of wireless sensor networks (WSNs). These WSNs use different sensor nodes with small battery power supply. Hence the energy of sensor node is considered as one of the primary constraints of WSN. In most WSNs applications, the sensed data which is generated from same location sensor nodes are identical or time-series/periodical data. This redundant data transmission leads more energy consumption. To reduce this energy consumption, a Data Reduction strategy using Neural Adaptation Phenomenon (DR-NAP) has been introduced to decrease the communication energy in routing data to the BS. Neural adaptation phenomenon has been utilized for designing a simple data reduction scheme to reduce the amount of data transmitted. By this way the sensor node energy is saved and the network lifetime is enhanced. The proposed approach has been implanted in the existing Gravitational Search Algorithm (GSA) based clustered routing for WSN. The sensed data is transmitted to CH and BS using DR-NAP. Real sensor data from Intel Berkeley Research lab has been used for conducting the experiments. The experiment results show 47.82% and 51.96% of improvement in network lifetime when compared with GSA based clustered routing and Clustering Scheme Using Canada Geese Migration Principle (CS-CGMP) for routing, respectively.

Index Terms—Wireless Sensor Networks, Internet of Things, Neural Adaptation, Energy Efficient Routing, WSN, Data Reduction, IoT

I. INTRODUCTION

Currently, Internet of Things (IoT) is growing technology along with the current scenarios such as wild-forest monitoring, medical research and healthcare, environmental monitoring, disaster relief, structural health monitoring, military, industry monitoring, etc. [1]–[3]. Most of the IoT applications are implemented with the help of wireless sensor networks (WSNs) [4], which are a collection of wireless sensing devices equipped with sensor nodes which are densely set-up to monitor the environment parameters such as humidity, pressure, temperature, vibration, acoustic sound, motion, pollutants, wind speed, wind direction, etc.

In WSN, energy efficiency is one of the challenging factor, which influences the design of network. There are numerous ways and means for attaining energy conservation in WSN. Energy-efficient routing, sleep/wakeup systems, battery replacement, radio optimization, and data reduction are the key methods suggested to tackle energy conservation [5].

Routing is an an important activity for communicating the sensed data to the Base station (BS) that consumes maximum energy. Hence routing protocols must focus on improving energy efficiency. Sensor nodes near by the BS are burdened since they need to forward lot of packets to BS and hence their energy exhausts soon in multi-hop communication. Various energy-saving mechanisms such as cluster architectures, optimal relay node placement, sink mobility, energy as a routing metric during setting up the path, and multipath routing can be adopted for efficient routing [6]. Another approach for addressing energy efficiency in WSN are energy harvesting and wireless charging technique to recharge the batteries of sensor nodes without human consultation [7]. In addition, another alternative solution to energy efficiency is Data Reduction, where reducing the amount of similar data to be communicated to the BS. Because data transmission is expensive in terms of energy to be spent.

The major contributions of the proposed research work are:

- Data redundancy reduction and transmission technique to reduce energy consumption
- Adoption of Neural Adaptation Phenomenon for data reduction in order to achieve energy efficiency
- Integration of proposed approach with GSA based clustered routing
- Network lifetime enhancement through reduced data communication.

Paper is prepared as follows: Section II describes Related work. Section III provides basics of Neural adaption phenomenon. Section IV describes the proposed data reduction techniques using nueral adaption phenomenon approach. The results of the experiments conducted are compared with the GSA based approach and CS-CGMP based approach in Sec-

tion V. Finally, the conclusion is presented in Section VI.

II. RELATED WORK

In cluster-based architecture, the network area is divided into a smaller section; each section is referred to as clusters. Each cluster possesses a chief, known as Cluster Head (CH). WSN Cluster-based architecture contributes in energy efficiency, scalability, reducing route-finding delay, and loop-free routing [8]. Various research works have been performed on energy-efficient cluster-based routing. The famous cluster-based routing strategies are LEACH (Low-Energy Adaptive Clustering Hierarchy) [9], DEEC (Distributed Energy-Efficient Clustering algorithm), HEED (Hybrid Energy-Efficient Distributed Clustering) [10] [11], etc.

In the literature, many load balanced cluster-based routing approaches [12]–[15] have been proposed for efficient routing. These approaches improve the network lifetime by achieving load balance among the CHs.

Cluster-based routing for WSN achieves energy efficiency; they still need improvement since energy conservation is vital for enhanced life of the network. To minimize the energy consumption of sensor nodes, several prediction strategies have been investigated [16]–[18]. These prediction strategies are performed based on the observation of the sensed data and sensor nodes are capable of performing computations for training and using predictors in a distributed manner. The representatives of the existing prediction strategies and data reductions methods are reviewed in this section.

MPEG encoding based prediction model [16] was proposed by Goel et al. to monitor the environment. This approach has taken the lessons from the MPEG encoding process for designing the prediction model. The prediction-model was sent to the sensor nodes after it was built. Only when the sensor nodes' sensed data differs from the reading given by the prediction model, or when the reading reaches a pre-defined threshold, do the sensor nodes send it to the monitoring devices. This phenomenon is called as PREDiction-based MONitoring (PREMON) paradigm. This paradigm avoids the unnecessary data transmissions from the sensor nodes by predicting the readings at the monitoring entity. However, extra computation has to be performed to determine the prediction model.

McConnell et al. [17] proposed a prediction strategy which is performed based on the targeted data. The sensor nodes transmit only the target data instead of sending all the sensed data. [18] proposed data-driven processing based on Spatio-temporal suppression strategy on continuous data but not continuous reporting of data. The authors adopted the models for optimized data collection. Ken [19] was suggested by Chu et al. as a robust approximation technique. To minimise data communication from sensor nodes to BS, this method used replicated dynamic probabilistic models.

Data compression methods for IoT sensor nodes have been presented using the Two-Tier Data Reduction (TTDR) technique [20]. The temporal correlation in the sensed data is used in this technique; Delta Encoding and Run-Length

Encoding(RLE) has been utilized for data compression. At the gateway tier, hierarchical clustering is used to group data sets from sensor nodes based on the Minimum Description Length (MDL) concept. The MDL theory compresses the data set pairs into a single group. As a result, the scale of the data sets is kept to a minimum.

A clustering-based data collection framework with a prediction scheme has been presented by Jiang et al. [21]. In this framework, a prediction scheme has been combined into data collection by adaptively enabling/disabling the prediction scheme. The adaptive scheme is adopted for controlling the prediction, and to investigate output tradeoff between between lowering communication costs and limiting prediction costs. In addition, for node-to-node propagation of aggregates, sleep/awake scheduling has been used; this framework makes use of an effective cluster-to-cluster propagation.

Jarwan et al. proposed a two-tier data reduction framework with Dual Prediction (DP) and Data Compression (DC) schemes [22]; spatio-temporal correlation is the basis for these schemes. To save energy and bandwidth, this framework reduces the number of data transfers. The DP scheme has been used to reduce transmissions between member nodes and CHs; the DC scheme has been used to reduce transmissions between sink nodes and CHs. However, Neural Network (NN) implementation of this framework is tedious in WSN because of the simple processing capability of sensor nodes.

Sayed et al. have introduced Distributed Data Predictive Model (DDPM) for decreasing the energy depletion of sensor nodes and thus enhancing the network lifetime [23]. For forecasting communication shortcomings, this strategy is based on a distributive clustering model. For removing unwanted reflections and transmitted signals with noise, a Recursive Least Squares (RLS) adaptive filter merged with a Finite Impulse Response (FIR) filter has been used; they seek to reduce the size of the data to be transmitted for energy efficiency.

In WSN, a prediction strategy based on the Hierarchical Least-Mean-Square (HLMS) adaptive filter [24] has been developed for data reduction. This filter was used by both the sensor nodes and the sink nodes to forecast measured values; as a result, sensor nodes could send sensor readings that vary by a certain amount from the prediction. The number of data sent by each node has been minimised in order to save energy.

A crucial issue in the conventional clustered routing is the redundant transmission of periodically sensed data which are alike for subsequent rounds of data transmission phase. Because transmission of identical data to the CH/BS is an energy-consuming process, it is considered a vital issue. Furthermore, the above-mentioned approaches involved the undue amount of computational overhead for performing the predictions and need to fix the error threshold correctly for efficient data prediction. To overcome the aforementioned limitations, a Data Reduction using Neural Adaptation Phenomenon (DR-NAP) dependent routing approach is proposed.

A. Neural Adaptation Phenomenon

Generally, a change in the habitat is the stimulus; the reaction of the organism to it is the response. Neural adaptation or sensory adaptation is defined as the receptivity of the sensory system is gradually decreasing to a constant stimulus. It is realized as a common phenomenon of dissolving neural activities concerning repeated or prolonged stimulation. It is a ubiquitous phenomenon and a dynamic process. It has been noticed in many neurons, periphery, and central nervous system (<https://psychologenie.com/understanding-sensory-adaptation-with-examples>).

Every phase of the stimulus, for example, a replay of pure accent of constant intensity or the deflection of a whisker by a fixed amplitude, causes any neuron responsive to the applied stimulus to respond with some simple response. A spiking neuron (non-adapting neuron) usually maintains a specific firing rate [25]. The adapting neuron lowers its response activity to a steady-state value.

Neural Adaptation are of two types: 1. Rapid Adaptation and 2. Slow Adaptation. Rapid adaptation evolves within a millisecond to seconds in the human body; it happens in the auditory system in the range of milliseconds and it evolves in many hundred milliseconds to seconds in the visual system. Slow Adaptation evolves in seconds to minutes, hours or days in the human body

This particular neural adaptation mechanism is related to data reduction in WSN sensor nodes. In most of the WSN applications, sensor nodes sense the same/similar data for a subsequent number of rounds. When the sensor node senses a data, it can immediately respond to its CH by transmitting the data. If the sensed value in the current round is the same as the sensed value in the previous rounds, it is not necessary to transmit data to the CH, instead sensor node may transmit a bit to indicate current value is equal to previous value. Otherwise, the sensor node transmits the currently sensed data to the CH. In this way, the number of data transmissions between the sensor node to CH can be reduced when the lifetime of the WSN is significant for running the system without any human intervention. Hence this intention inspired us to adopt a neural adaptation phenomenon for data reduction in the routing of WSN.

III. PROPOSED APPROACH

The proposed approach used cluster communication for collecting the data from different sensor based IoT devices. Clustering & communication process consists of two phases:

- 1) Clustering Phase - This phase includes partitioning the network field into clusters, choosing a precise number of CHs, and binding the sensor nodes to an appropriate CH.
- 2) Data Transmission Phase - This phase includes communicating data to the CH with the aid of neural adaptation phenomenon and later forwarding the data from CH to BS.

A. Clustering Phase

The network field is divided into two layers: the border layer and inner layer. In the proposed approach, ten percent of the network area in the boundary is considered as border layer; the remaining area is referred to as an inner layer. Ten percent of the network area denotes ten percent in the height and width of the area. As the boundary nodes are far away from the BS, their data communication distance is more and hence the boundary nodes exhaust energy soon. Accordingly, the sensor nodes in the border layer are not considered for the cluster head selection. The CHs are selected from the sensor nodes in the inner layer; the sensor nodes in the border layer can communicate the data through the nearest CHs in the inner layer. The inner layer is sub-divided into squared zones. The node density of a zone is measured as the total number of nodes which are alive in a zone. The clustering phase is repeated for every stipulated number of rounds.

1) *Selection of CH*: CHs are very important for cluster based communication in WSN. CH selection phase is performed as in CS-CGMP based routing strategy [13]. The number of CHs will be elected with the help of node density in a network as well as in zone. The node density of the zone will decide the number of CHs.

Computing number of CHs: The required number of CHs for every zone is computed using Equation 5. It is determined based on the node density of the zones.

$$Num_CH_{Z_i} = round\left(\frac{Node_Density_{Z_i}}{AVG_NODE_DENSITY}\right) \quad (5)$$

where $Node_Density_{Z_i}$ denotes number of nodes in i^{th} zone, $AVG_NODE_DENSITY$ is average number of nodes present in entire network by considering all the zones and is calculated using $\left(\frac{Total_Number_of_Nodes}{Number_of_zones}\right)$. Node which is having more energy compared to other nodes and considering the overall communication and transmissions distances with all other sensor node and with BS will be considered for selection of a CH.

2) *Cluster Set-up Phase*: In this phase, GSA-based clustering approach [13] has been utilized for the balanced binding of sensor nodes in the inner layer to the CHs after selecting the required CHs. BS determines the best binding of a sensor node to CH based on residual energy of CH, energy to be spent for its member nodes for aggregation by CH, and sensor node to CH distance. The sensor nodes in the border layer are associated with the nearby CHs. After finding the optimal binding of sensor nodes to the appropriate CHs, BS sends a short message to all sensor nodes. The message for the member sensor node consists of an assigned CH_ID . Further, the message for the CHs consists of a list of member sensor nodes and its forwarding CH.

B. Data Transmission Phase - Neural Adaptation Phenomenon

The data transmission phase includes two sub-phases:

- 1) Member sensor nodes to CH Communication
- 2) CH to BS Communication.

Each sensor node sense the data and it can store value. Second time onwards it will sense and compare with stored value, if they are same then transmit flag value is set to 0. Instead of transmitting sensed value, sensor node will transmit the flag, which indicates previous value.

1) *Member sensor nodes to CH Communication:* In this phase, the neural adaptation phenomenon has been utilized to enhance the network lifetime by reducing the number of data transmissions. After clusters are formed, the member sensor nodes can send the sensed data to respective CH during its allotted slot. Sensor nodes may sense the same/similar data for subsequent rounds in periodical data transmission. If sensed value matches with the previous value then sensed data is not transmitted instead a flag will be transmitted to indicate same as previous value or no change in data compared to previous value. Otherwise, sensed data will be transferred from sensor node to the CH. When CH receive a flag bit then it takes the previously received data as the current value. Hence, the number of data transmissions between the sensor node to CH has been reduced. Then the CHs will aggregate the data which receives from sensor nodes and then forwards to the BS.

2) *CH to BS Communication:* When a CH receive data from its member sensor nodes for every round of data transmission, it will store the received value as previous value. When a CH receives a flag (which indicates same as previous sensed value), it will take previous value as current value.

When CH has not received any data, *Nodata_Count_i* is incremented. If *Nodata_Count* is greater than *Nodata_Thresh*, the member node is declared as dead node. If the *Alike_Flag* is set, CH takes the previous value as the current value. If no flags are set, the two previous values are same then CH takes the previous value as current value. Then *Alike_Flag* is set; otherwise it is considered as sequence. Then current value is calculated from the two previous values; *Sequence_Flag* is set. Similar procedure is applied for all the member nodes. Finally, using multi-hop communication, CH aggregate and communicate the data to BS. CHs will transmit its aggregated data to the BS through intermediate CH, which is identified with the help of residual energy and distance to BS.

IV. SIMULATION RESULTS AND DISCUSSION

The simulation experiments of the proposed approach were conducted to validate the performance of data reduction strategy DR-NAP; it has been incorporated and implemented with GSA-based Clustering Routing Strategy (GSA-CRS) [13] and Clustering-Scheme using Canada Geese Migration Principle (CS-CGMP) [26]. The results of the proposed approaches were considered and compared with no data reduction strategy techniques.

WSN scenario as shown in Fig. 1 is used by Intel Berkeley Research Lab (<http://db.csail.mit.edu/labdata/labdata.html>) and generated the data as shown in Fig. 2 has been considered for simulation. In this scenario, 54 sensor nodes were deployed on $44m \times 35m$ network field with BS placed at the center. For

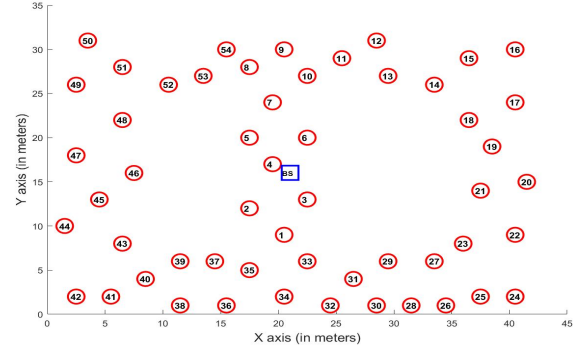


Fig. 1: WSN Scenario - Intel Berkeley Research Lab

the simulation, sensor readings of Intel Berkeley Research lab between February 28th and April 10th, 2004 was used. Mica2Dot sensor nodes collected temperature, humidity, voltage, and light values, along with topology information for every 31 seconds.

A. Result Analysis

In this section, the performance is compared between GSA-CRS and CS-CGMP approaches with/without DR-NAP is presented. To verify the performance, lifetime and remaining energy of network are considered in this paper. Fig. 3 and Fig. 4 depicts the performance comparison of GSA-CRS and CS-CGMP approaches with/without data reduction.

Network lifetime: In DR-NAP, the number of data communications have been reduced and hence reduces the communication cost. Further, the network lifetime of GSA-CRS with DR-NAP is improved by 47.82% when compared with GSA approach without DR-NAP. The network lifetime of CS-CGMP approach with DR-NAP improved by 51.96% when compared with CS-CGMP approach without DR-NAP. Fig. 3 shows the network lifetime of proposed approaches namely GSA-CRS, CS-CGMP with and without Data reduction for temperature data.

Average Residual energy of Node: Average residual energy of every node in the network over the rounds of 2300 is shown in Fig. 4. The proposed approaches GSA-CRS with DR-NAP, CS-CGMP with DR-NAP have improved their energy level on average, by 50.75% and 40.87% more than GSA-CRS and CS-CGMP without DR-NAP respectively. The number of data packet transmission is reduced which leads in reduction of energy consumption and residual energy usage.

V. CONCLUSION

A neural adaptation phenomenon based data reduction technique for Internet of Things and Wireless Sensor Networks has been proposed in this paper. The DR-NAP strategy has been implanted in GSA based clustered routing for reducing the number of data transmissions between sensors to CHs and BS. Member sensor nodes will communicate the sensed data to the CHs based on the recently sensed values. If the recently sensed readings are identical or in sequence, sensor nodes

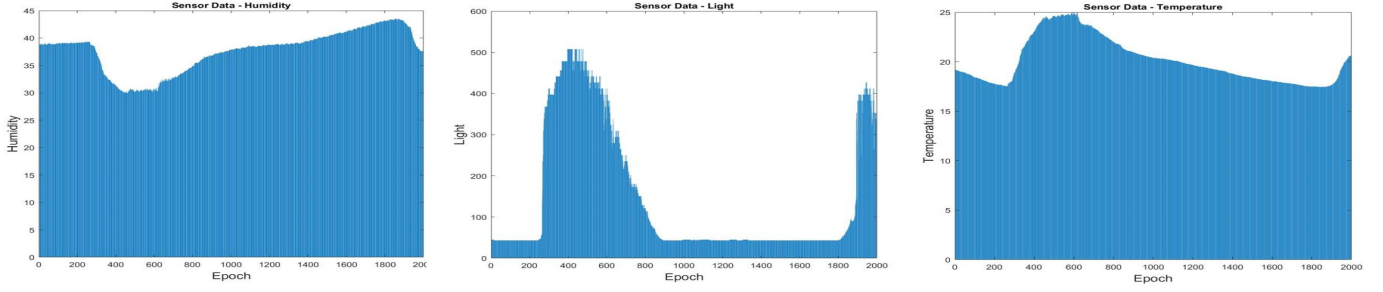


Fig. 2: Sample of Intel Berkeley Research Lab's sensed data (<http://db.csail.mit.edu/labdata/labdata.html>)

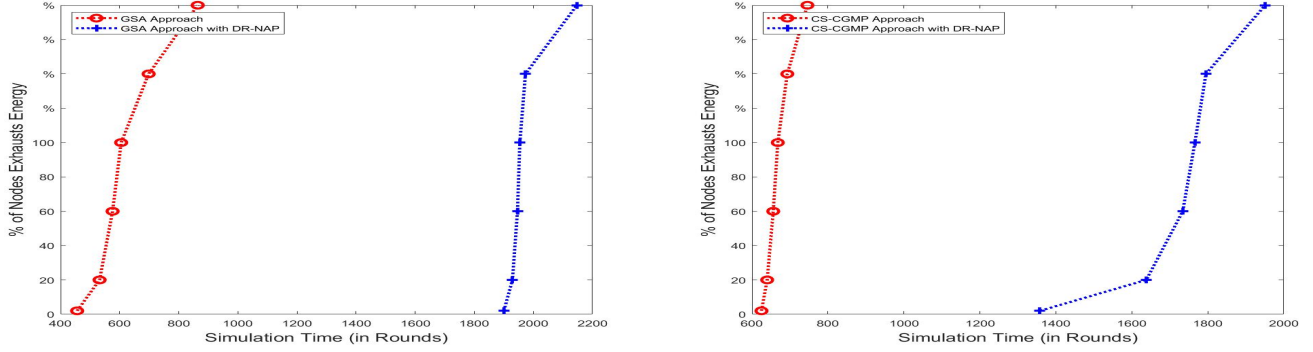


Fig. 3: Comparison of Network lifetime for GSA-CRS, CS-CGMP approaches with and without Data reduction

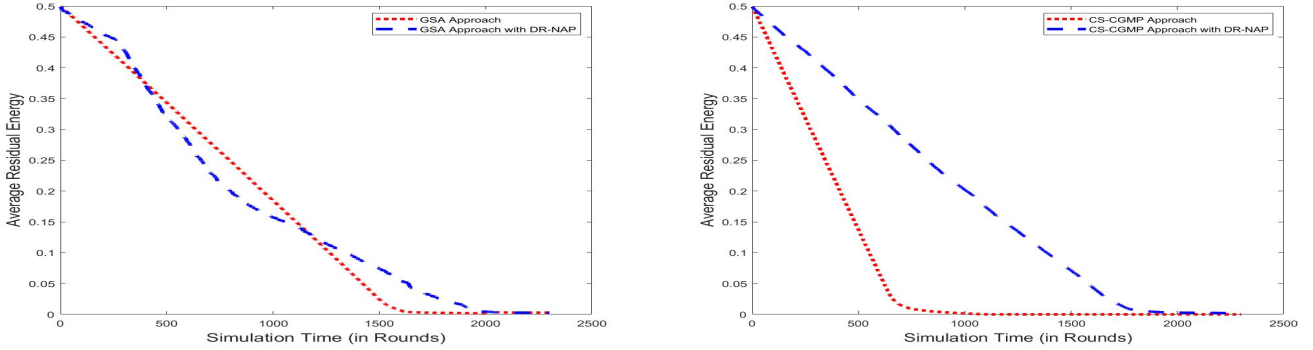


Fig. 4: Comparison of Average Residual Energy for GSA-CRS, CS-CGMP approaches with and without Data reduction

need not transmit, instead it may send an indicator signal bit 1 stating that current recorded value is equal with previous round value; otherwise, the sensed valued are transmitted. Hence, the total energy consumption has been decreased in individual sensor nodes. As per experiments conducted, it was noticed that the performance parameters such as network lifetime and average residual energy have been improved. The efficacy of GSA-CRS with DR-NAP has been enhanced by 47.82% in network lifetime when compared with GSA-CRS without DR-NAP; the CS-CGMP with DR-NAP has enhanced its network lifetime by 51.96% against CS-CGMP without DR-NAP. In future research works, the proposed data reduction strategy can be extended for handling multimedia data like image, video, audio, etc. in Wireless Multimedia Sensor Networks (WMSN)

and IoT.

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